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In the above program, after getting the values of num1 and num2, sum is declared and initialized with the addition of those two variables. After that, it is divided by 2.

Illustration 9.8: C++ program to find the perimeter and area of a semi circle

```
#include <iostream>
using namespace std;
int main()
{
        int radius;
        float pi = 3.14;
        cout << "\n Enter Radius (in cm): ";
        cin >> radius;
        float perimeter = (pi+2)*radius;   // dynamic initialization
        float area = (pi*radius*radius)/2; // dynamic initialization
        cout << "\n Perimeter of the semicircle is " << perimeter << " cm";
        cout << "\n Area of the semicircle is " << area << " sq.cm";
}
```

Output:

```
Enter Radius (in cm): 14
Perimeter of the semicircle is 71.96 cm
Area of the semicircle is 307.72 sq.cm
```

9.13.4 The Access modifier const:

const is the keyword used to declare a constant. You already learnt about constant in the previous chapter. const keyword modifies / restricts the accessibility of a variable. So, it is known as Access modifier.

For example,

int num = 100;

The above declares a variable num with an initial value 100. However, the value of num can be changed during the execution. If you modify the above definition as **const int num = 100;** the variable num becomes a constant and its value will remain as 100 throughout the program, and it can never be changed during the execution.